2.5.8 Documenting *Mathematica* Programs

f`::usage = "`*text*`"`	text about a function
`(*` *text* `*)`	comment within a function

Ways to document functions in *Mathematica*.

When you write a function in *Mathematica* that you intend to use in the future, it is a very good idea to put in some kind of documentation that tells you about the function. If you define a value for *f*`::usage`, this will be retrieved when you type `?`*f*.

Here is the definition of a function `f`.

```
In[1]:= f[x_] := x^2
```

Here is a "usage message" for `f`.

```
In[2]:= f::usage = "f[x] gives the square of x."
Out[2]= f[x] gives the square of x.
```

This gives the usage message for `f`.

```
In[3]:= ?f
f[x] gives the square of x.
```

`??f` gives all the information *Mathematica* has about `f`, including the actual definition.

```
In[3]:= ??f
f[x] gives the square of x.
            2
f/: f[x_] := x
```

When you define a function *f*, you can usually display its value using `?`*f*. However, if you give a usage message for *f*, then `?`*f* just gives the usage message. Only when you type `??`*f* do you get all the details about *f*, including its actual definition.

If you ask for information using `?` about just one function, *Mathematica* will print out the complete usage messages for the function. If you ask for information on several functions at the same time, however, *Mathematica* will just give you the name of each function.

If you use *Mathematica* notebooks, you can give complete documentation for your functions in text cells that surround the cells where you give the definitions.

Whether you are using notebooks or not, you can include text in the form of "comments" in your *Mathematica* input. Any text that you put between `(*` and `*)` will be ignored when a *Mathematica* expression is evaluated. (You can nest `(*` and `*)`.)

You can use comments anywhere in your input.

```
In[3]:= If[a > b, (* then *) p, (* else *) q]
        (* This is a conditional *)

Out[3]= If[a > b, p, q]
```

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